

# Pramaan — EPC Deviation Intelligence

Spec-to-Site Deviation Sentinel + Commissioning Risk Twin · ET AI Hackathon 2026 · Problem Statement 4

Repository: [github.com/bansalbhunesh/parth](https://github.com/bansalbhunesh/parth) · Judge Mode: [parth-tan.vercel.app/judge](https://parth-tan.vercel.app/judge) · Evidence: [parth-tan.vercel.app/evidence](https://parth-tan.vercel.app/evidence) · API: [parth-1-ma30.onrender.com/health](https://parth-1-ma30.onrender.com/health)

Phase-2 deadline (live Unstop record): **2026-07-22, 23:59:21 IST**

**The wedge:** Pramaan converts a spec/submittal mismatch into a cited commissioning failure mode and the schedule window to act — *before* the failed test becomes a project delay. The trace is owner requirement → deviation → commissioning test → lead-time-to-failure, with evidence at each step. It is a **benchmarked prototype**, not a field-validated product, and this document labels every number accordingly.

**Evidence labels used throughout:** **BENCHMARKED** from the frozen ps4\_external\_v1 (v1.2) run **DETERMINISTIC OFFLINE** reproducible with no API key **LIVE-MODEL** varies run-to-run **SYNTHETIC** authored corpus, true by construction **SCENARIO** modeled assumption, not validated **PENDING** not yet done.

## 1. Problem statement & PS4 alignment

In a hyperscale data-centre EPC build, the **design basis**, **vendor submittals**, and **governing standards** live in different systems and are reviewed by different people. A vendor deviation missed at submittal review resurfaces months later at commissioning — as a failed acceptance test, rework, and schedule slip. Found on submittal day it is a one-line RFI; found at commissioning it is a schedule event. PS4 asks for AI that reconciles specifications against submittals and standards; Pramaan does that and adds the commissioning/schedule consequence and the window to act.

Market context **SCENARIO**: India data-centre capex is large and growing (secondary research). These frame the opportunity; they are not Pramaan-specific results.

## 2. Product workflow

- Reconciles a design specification against a vendor submittal and the governing standards, and reports each genuine deviation with the standard it violates and a citation chain (required value, submitted value, source, confidence).
- For each deviation, predicts the **commissioning test** that will fail and the **lead-time-in-weeks** to fix it if caught now — the commissioning-risk twin — and connects it to schedule and supply-chain services.
- Judge Mode ( /judge )** is the 60-second flow: *Load deviation demo* ★ (a natural-prose pair with a hidden 2N→N+1 and 10→8 min non-compliance) and *Load compliant demo* ✓ (a fully compliant submittal — the correct answer is zero deviations). Users can upload their own spec + submittal (PDF / image / text).
- Every result shows a **truthful provenance chip** — live LLM, vision, deterministic rule floor, or OCR — and a zero-deviation result from the rule floor is labelled inconclusive, not a green pass.

## 3. Technical architecture (as actually executed)

Pramaan is **one compliance reasoning graph, a set of connected deterministic intelligence services, and a reliability layer** — *not* five co-operating LLM agents. Verify against `backend/orchestrator.py`. The runtime graph is:

`ingest → load_standards → validate → reconcile → retrieve → critique → cx_predict → format_output`

- Exactly one node reasons with an LLM** — `reconcile` (extraction happens inside it; there is no separate extract node). `ingest`, `load_standards`, `validate`, `retrieve`, and `critique` are deterministic.

- Two **bounded cycles**: a retrieval tool-call (`retrieve` fetches a cited standard missing from context and loops back) and a self-critique / reflexion loop (`critique` drops equality-false-positives and duplicates, loops back on a failed check). Both are bounded by env vars; a sequential runner drives the same nodes if LangGraph is absent.
- `cx_predict` maps each deviation to its commissioning test via a rule table then a standards graph; it calls an LLM only as a **fallback** for equipment classes in neither.
- **Connected deterministic services** (separate endpoints, not graph nodes): schedule risk (graph CPM + beta-PERT Monte Carlo), supply-chain risk (closed-form ETA/risk), and the project/Cx graph. The **RFI Copilot is a service** (BM25 retrieval + an LLM only to phrase a cited answer), *not* a graph node.

## 4. Benchmark methodology & results

The headline evaluation is the frozen, provenance-tracked **ps4\_external\_v1 (v1.2)** benchmark: **53 spec-submittal pairs, 129 single-author-frozen labels, 17 datacenter system types, and 64 clean-negative (compliant) controls**. The featured configuration is `gemini-3.1-flash-lite` over a 3-pass repeated run. Source: `benchmarks/ps4_external_v1/reports/benchmark_card.json` (regenerate with `python scripts/benchmark_report.py`).

### RECALL **BENCHMARKED**

**0.862** mean semantic recall  
(range 0.841–0.873 across 3 passes).

### PRECISION **BENCHMARKED**

**0.953** mean semantic precision.

### F1 **BENCHMARKED**

**0.905** mean semantic F1.

### FALSE-ALERT RATE **BENCHMARKED**

**0.000** on the 64 clean-negative controls.

### LATENCY **BENCHMARKED**

**~2.5 s** p50 per pair  
(gateway/model dependent).

### RULE BASELINE **DETERMINISTIC OFFLINE**

**0.111** recall (7/63), 0 false positives — a deliberately low floor.

The scorer performs **one-to-one (bipartite) matching** — each finding is credited to at most one label and vice-versa, so a broad finding cannot inflate recall, and unmatched findings count as false positives. Recall counts not-run (quota/outage) pairs as misses. The reasoning core clears the deterministic floor on the same frozen labels (0.862 vs 0.111).

### Provenance & review status (the limitations that must travel with the numbers).

- Fixtures are **team-authored**; **10** are derived from public primary sources (**5** with a verified public URL).  
**SOURCE FILES NOT STORED YET**
- Labels are **single-author frozen**. A second human reviewer (reviewer-2) has **not** run:  
**TWO-PERSON ADJUDICATION PENDING**
- An automated consistency audit checks each label against its own documents (123/129 consistent, 6 flagged). This is **machine QA, not a human reviewer**, and does not substitute for the pending adjudication.
- This is a **benchmark result** — **not** a real-world-accuracy, field-validation, or customer claim.

A comparison model (`gemini-2.5-flash`) reached a higher peak recall (~0.95) but was slower and did not complete a clean 3-pass repeat; it is reported as an ablation, not the primary result. Full card and live status: `parth-tan.vercel.app/evidence`.

## 5. OCR, failover & reliability

### PROVIDER FAILOVER MEASURED

gemini → Qwen gateway →  
Groq → Claude → local Ollama  
→ deterministic rule floor. Only  
configured providers are tried.

**Availability, not accuracy** —  
every leg is scored the same.  
Live chain at `/llm-check` (?  
deep=1 for a reconcile-sized  
probe).

### DETERMINISTIC FLOOR DETERMINISTIC OFFLINE

No silent zeros: if no provider  
answers, the rule engine returns  
headline deviations computed  
from the documents (low-recall  
by design, 0.111), never seeded  
labels. The UI labels a rule-floor  
0-result as inconclusive.

### OCR MEASURED

Reads scanned / image-only  
PDFs and uploaded images via  
Tesseract (text-first, best-effort).  
`render.yaml` builds from  
`Dockerfile.backend` (Tesseract  
in-image); `GET /ocr-check`  
reports live status. Hosts  
without Tesseract return a clear  
"OCR unavailable" message.

`/health` exposes the running commit and analysis mode; every endpoint returns 200 without a key (bundled ground-truth data), so a cold or key-less backend never fakes a live result.

## 6. Security & public-demo hardening (implemented)

Phase-5 shipped **demo-level** hardening (verified by `tests/test_security*.py`, `test_upload_hardening.py`, `test_prompt_injection.py`, `test_no_secrets.py`):

- **Optional token auth** on analysis endpoints and **per-IP rate limiting** (single-instance, in-memory).
- **Upload validation** — MIME + extension + magic-byte allowlist; rejects archives/executables/disguised files, oversized uploads (15 MB cap), and decompression-bomb images.
- **Prompt-injection-resistant** prompts and **no secret leakage** in status endpoints (booleans/caps only; the token and client IP are never returned).

**Demo hardening, not production security.** Rate limiting has no shared store, auth is an optional demo token (not access control / tenant isolation), and this is not a production-load story. These are named limitations, not claims. See `docs/SECURITY_DEMO_RUNBOOK.md`.

## 7. Standards, IP & provenance limitations

- Third-party standards content is a team-authored paraphrase from public/secondary references — proprietary standard text is not redistributed; product and standard names are trademarks of their owners, and Pramaan is not endorsed by them.
- Benchmark provenance is governed by `docs/CLAIMS_REGISTER.md` and `docs/SOURCE_PROVENANCE.md`: allowed vs banned wording, each quantitative claim paired with its limitation. "Derived from a public primary source" is not the same as a stored primary-source PDF, and we do not conflate them.
- No API keys are committed; credentials are external secrets. `THIRD_PARTY_NOTICES.md` records third-party artifacts.

## 8. Deployment & reproducibility

### TESTS MEASURED

**888** automated tests (API,  
agents, security, resilience,  
benchmark, failover) + 155

### REPRODUCE OFFLINE DETERMINISTIC OFFLINE

python  
`scripts/run_benchmark_suite.py`  
`--rule-only` (manifest + hash +

### DEPLOY MEASURED

Frontend on Vercel, API on  
Render (Docker backend).  
`/health` shows the running

Playwright E2E journeys. Full CI:  
Python tests + lint + pip-audit  
+ bandit, frontend  
types/tests/build/E2E, Docker.

rule baseline + report) — no API  
key, a judge's laptop is enough.

commit; `verify_live.py`  
checks the deployment end-to-  
end.

Reproduce the benchmark card: `python scripts/benchmark_manifest_check.py`, `benchmark_hash_sources.py`,  
`benchmark_report.py`. Deterministic evals and unit tests run offline; live-model results are labelled and vary run-to-run.

## 9. Known limitations & roadmap

- **Benchmark:** team-authored fixtures, single-author frozen labels; source-file archiving and two-person (reviewer-2) adjudication are **PENDING**. Not field- or customer-validated.
- **Schedule risk is simulation**, not field-calibrated prediction; deviation probabilities are seeded, not empirical. Supply-chain intelligence is closed-form over static data (no live feed).
- **No measured coordination-hours study** yet; any cost/ROI figure is an explicitly-labelled scenario, not realized savings.
- **Prototype hardening:** single-instance in-memory rate limiting, optional demo token; synchronous LLM calls; not a production-load service.
- **Next milestones (trust-first):** archived source artifacts + expanded primary-source-derived pairs; reviewer-2 two-person adjudication and resolving the 6 audit-flagged labels; production hardening (shared-store auth/rate-limit, pgvector/Qdrant, async, delta re-checks).

## 10. Links

- **GitHub:** [github.com/bansalbhunesh/parth](https://github.com/bansalbhunesh/parth)
- **Live demo — Judge Mode:** [parth-tan.vercel.app/judge](https://parth-tan.vercel.app/judge)
- **Evidence dashboard:** [parth-tan.vercel.app/evidence](https://parth-tan.vercel.app/evidence)
- **API health / OCR / LLM status:** [/health](#) · [/ocr-check](#) · [/llm-check](#)
- **Benchmark report + card:** [benchmarks/ps4\\_external\\_v1/reports/benchmark\\_report.md](#) · [benchmark\\_card.json](#)
- **Deck (PDF):** [docs/Pramaan\\_Deck.pdf](#) · **Claims register:** [docs/CLAIMS\\_REGISTER.md](#)
- **Demo video:** **LINK PENDING — SEE DOCS/VIDEO\_RUNBOOK.MD**

## Rubric mapping (PS4 weights: 25 / 25 / 20 / 15 / 15)

| Criterion                  | How Pramaan addresses it                                                                                                                                                                | Honest status                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Innovation (25%)           | Spec → deviation → commissioning test → lead-time chain; a distinctive wedge beyond submittal review                                                                                    | Strong — the moat is the Cx-test / lead-time link, not module count                                                     |
| Business impact (25%)      | Cost-of-late-nonconformance thesis; expected-value impact model (docs/BUSINESS.md) with low/base/high assumption bands, not a single deterministic number                               | <b>SCENARIO</b> base case ~3.2x on modeled assumptions, honestly negative in the low case; no paid-pilot validation yet |
| Technical excellence (20%) | LangGraph reasoning graph + one LLM core, deterministic services, 888 tests, CI, frozen benchmark                                                                                       | <b>BENCHMARKED</b> 0.862/0.953/0.905; reviewer-2 pending                                                                |
| Scalability (15%)          | Modular, containerized; deterministic services run at high throughput; async job flow + input-hash cache; persisted case workflow (SQLite, tenant-isolated) for submittal→RFI→audit-log | Prototype: single-instance auth/rate-limit; LLM path is the scale constraint                                            |
| User experience (15%)      | Focused Judge Mode + streaming reasoning + evidence page; provenance chips                                                                                                              | Lighthouse (/judge): Performance 53→92, Accessibility 92→100 this phase                                                 |

## Anticipated judge questions (honest answers)

**Is the data real?** The benchmark fixtures are team-authored — 10 derived from public primary sources, 5 with a verified URL — and source files are not stored in this benchmark yet. We do not call them real datasheets. The benchmark measures a reliable first-pass workflow, not real-world accuracy.

**Is the benchmark independently reviewed?** Not yet. Labels are single-author frozen; the automated consistency audit (123/129) is machine QA, not a second human reviewer. Two-person adjudication is the next validation milestone.

**Why not "five agents"?** Exactly one node reasons with an LLM (`reconcile`); ingestion, retrieval, self-critique, and commissioning mapping are deterministic and inspectable, plus deterministic schedule/supply/graph services. One reasoning graph + deterministic services, not five agents.

**What if the LLM is down during judging?** Every endpoint returns 200, the rule engine still finds headline deviations, and the UI labels that as the low-recall floor — not a clean bill of health. `/llm-check` reports the true status.

Evidence-safe proof points: "On the frozen ps4\_external\_v1 (v1.2) benchmark — 53 team-authored pairs, 129 single-author-frozen labels, 17 systems, 64 clean negatives — the featured 3-pass run reports recall 0.862, precision 0.953, F1 0.905, and 0 false alerts on the clean negatives, vs a deterministic rule baseline of 0.111." · "888 automated tests; full CI." · "Reviewer-2 human adjudication is pending; the automated cross-check is machine QA." · Every metric carries an evidence label. Results are observations at the audited commit, not evergreen badges.